# Correction

## EVOLUTION

Correction for "Causes of natural variation in fitness: Evidence from studies of *Drosophila* populations," by Brian Charlesworth, which appeared in issue 6, February 10, 2015, of *Proc Natl Acad Sci USA* (112:1662–1669; first published January 8, 2015; 10.1073/pnas.1423275112).

The authors note that on page 1666, right column, first full paragraph, line 6, "an estimate of $\bar{t}$ of 0.044" should instead appear as "an estimate of $\bar{t}$ over both classes of mutation of 0.0073."

The authors also note that on page 1666, right column, fourth full paragraph, line 3, "although even this is likely to be a substantial underestimate" should instead appear as "although the overall value is likely to be somewhat less than 0.01."

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CORRECTION

# Causes of natural variation in fitness: Evidence from studies of *Drosophila* populations

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Contributed by Brian Charlesworth, December 12, 2014 (sent for review September 18, 2014; reviewed by James Fry and David Houle)

**DNA sequencing has revealed high levels of variability within most species. Statistical methods based on population genetics theory have been applied to the resulting data and suggest that most mutations affecting functionally important sequences are deleterious but subject to very weak selection. Quantitative genetic studies have provided information on the extent of genetic variation within populations in traits related to fitness and the rate at which variability in these traits arises by mutation. This paper attempts to combine the available information from applications of the two approaches to populations of the fruitfly *Drosophila* in order to estimate some important parameters of genetic variation, using a simple population genetics model of mutational effects on fitness components. Analyses based on this model suggest the existence of a class of mutations with much larger fitness effects than those inferred from sequence variability and that contribute most of the standing variation in fitness within a population caused by the input of mildly deleterious mutations. However, deleterious mutations explain only part of this standing variation, and other processes such as balancing selection appear to make a large contribution to genetic variation in fitness components in *Drosophila*.**

mutation | selection | genetic variability | *Drosophila*

Advances in DNA sequencing methods have enabled geneticists to measure the amount of genetic variability in natural populations at the most basic level: the frequencies of variants in nucleotide sequences. This achievement has ended one component of a debate on the extent and causes of genetic variability that was initiated in the 1950s by Hermann Muller and Theodosius Dobzhansky (1, 2); we now know that DNA sequences are highly variable within the populations of most species (3). It has, however, been much harder to provide a definitive answer to the other component of this debate, which concerns the nature and intensity of the evolutionary forces that influence the frequencies of genetic variants within populations (1, 2, 4, 5). Are these variants mostly selectively neutral (6), with the fates of new mutations determined by random fluctuations in their frequencies (genetic drift)? Is selection on variants that affect fitness mostly purifying, so that mutations with harmful effects are rapidly removed from the population (1)? Or do many loci have variants maintained by balancing selection (2)? What fraction of newly arisen variants cause higher fitness and are in the process of spreading through the population and replacing their alternatives? How strong is the selection acting on nonneutral variants, and how much variation in fitness among individuals within populations is contributed by such variants? Does the existence of wide variation in fitness among individuals imply a genetic load that threatens the survival of the species (1)?

These questions are very broad, and this paper deals only with one aspect of them. It focuses on the question of how recent inferences concerning the strength of purifying selection, derived from genome-wide surveys of DNA sequence variability, can be connected with the results of statistical studies of genetic variation in components of Darwinian fitness such as viability and fertility. I will refer to these two approaches as population genomics and quantitative genetics, respectively. The first approach sheds light on the general nature of the fitness effects of the DNA sequence variants found in natural populations, but says little about how these fitness effects are caused. The second tells us how much genetic variability exists for fitness traits, the rate at which it arise by mutation and something about the type of selection involved, but is silent about the nature of the underlying sequence variants.

Surprisingly little attention has been paid to integrating these two lines of inquiry, except for ref. 7. I largely confine myself to results from studies of the fruitfly *Drosophila*, because this has been the most useful model organism for investigating these problems, especially by quantitative genetics methods. Current information derived from population genomics studies will first be reviewed, followed by an analysis of the results of quantitative genetics experiments on both mutational and standing variation. I show that the quantitative genetics results can only be explained if there is a significant input of new mutations with much larger effects on fitness than those inferred from population genomics. There also appears to be too much genetic variation in fitness components in natural populations to be explained purely by mutation selection balance, so that additional processes such as balancing selection must make an important contribution.

## Population Genomics Analyses

DNA sequencing has revolutionized studies of three aspects of variability: the mutational processes that generate new variants, the amount of variation between individuals within a species, and the extent of between-species differences. Analyses of the evolutionary forces affecting the fates of DNA sequence variants need to use all three types of information. Here I discuss evidence about

### Significance

**The extent and causes of genetic variation have been debated for 60 y. This paper synthesizes evidence from studies of DNA sequence variability in *Drosophila* and from experiments on the quantitative genetics of fitness components. Two major conclusions emerge. First, a class of mutations with relatively large fitness effects contributes importantly to both the overall effect of new mutations on fitness components and to standing variation. These mutations are not detected in analyses of sequence variability. Second, a large fraction of variability in fitness components must be maintained by selection rather than reflecting deleterious variants introduced by mutation. These results imply that both approaches to the study of natural variation are needed to obtain a complete understanding of its causes.**

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See Profile on page 1652.

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the rate at which selectively deleterious mutations arise in the genome as a whole (the "deleterious mutation rate"), and the extent to which sequence variants affect fitness.

**Mutation Rates and Levels of Selective Constraint.** It is now possible to estimate the rate of mutation to nucleotide sequence variants for the genome as a whole, by determining the rate at which these arise de novo in families or in breeding experiments, as described below. We can also estimate the extent to which purifying selection acts to eliminate deleterious mutations that arise in different compartments of the genome (coding sequences, intron sequences, and intergenic sequences), as indicated by their relative degrees of sequence divergence from a related species (8). The proportion of sites that are conserved for a given class of sequence, relative to a putatively neutral standard such as inert transposable element sequences, measures the level of selective constraint for this class (9, 10). The deleterious mutation rate can then be estimated by multiplying the total numbers of sites in the different classes by their levels of selective constraint and then multiplying the sum of these products by the mutation rate per nucleotide site (8). A somewhat different approach, based solely on within-species variation, has also been used (11, 12), but the general principle is similar.

This procedure estimates the number of new mutations that occur in the genome of a newborn individual and that are so strongly selected against that they are virtually certain never to spread through the species by genetic drift; this requires the product of the strength of selection on such mutations and the effective population size $N_e$ (12, 13) to be >1 (14, 15). Because measures of DNA sequence variability indicate that $N_e$ is usually at least several thousand (3), reaching a million or more for many *Drosophila* species, the deleterious mutation rate estimated in this way includes contributions from mutations that reduce fitness by 0.1% or less.

Several DNA sequence analyses of mutation accumulation (MA) experiments have been performed on the *D. melanogaster* nuclear genome (12, 16, 17). These experiments used sets of sibmated lines derived from a single inbred ancestral stock maintained for many generations by brother-sister mating, so that all but highly deleterious new mutations are rapidly fixed within a given line (18). Combining these estimates of mutation rates with the estimates of selective constraints suggests a value of ~0.5 for the deleterious mutation rate of a *D. melanogaster* haploid nuclear genome, caused by base substitutions plus small insertions and deletions (12, 16, 17). However, this estimate is subject to considerable uncertainty, because inbred lines with independent origins differ in their mutation rates (12, 16, 17). The mutation rate for a typical outbred fly is therefore not known with much confidence. A value of 0.5 will be used in the discussions below, but should be regarded as provisional. Large-scale experiments using pairs of outbred individuals and their offspring are needed to obtain a more reliable estimate (19).

**Estimating the Fitness Effects of Mutations from Polymorphism Data.** The availability of DNA sequences from multiple individuals of the same species, for sets of many genes or for whole genomes, has motivated the development of methods for inferring the type of selection acting on nucleotide variants and estimating the strength of such selection by means of statistical fits of population genetics models of mutation, genetic drift and selection (20–29). Most applications of these methods have analyzed selection on nonsynonymous variants, with the assumption that these mostly reflect mutations that are either neutral or slightly deleterious (28), consistent with the fact that the majority of nonsynonymous variants are present at low frequencies.

These procedures assume that new deleterious mutations entering the population follow a probability distribution of the selection coefficient $t_i$ that describes the strength of selection on a heterozygous, nonsynonymous new mutation at a random site $i$ in the genome: this is the distribution of fitness effects (DFE) (Box 1). Most deleterious mutations remain rare in the population, so that selection on homozygotes has little influence on their fate in a randomly mating population; $t_i$ is thus the most relevant measure of the effect of a mutation on fitness. The largest *Drosophila* dataset thus far analyzed is that from a Rwandan population of *D. melanogaster*, for which next-generation whole genome sequencing data from multiple haploid genomes is available (29, 30). A log-normal distribution of $t_i$ provided the best fit, such that the mean of $N_e t_i$ was about 680, where $N_e$ was estimated as $0.7 \times 10^6$ from the level of synonymous site diversity (29). The estimated mean of $t_i$, $\bar{t}$, is ~0.001. The shape parameter of the distribution [the exponential of its standard deviation (SD)] was 4.9, which corresponds to a coefficient of variation of $t$ ($CV_t$) of 2.0.

Because of its size, this dataset has much greater statistical precision than previous studies of *Drosophila* populations, but the overall pattern is broadly similar to those reported previously (23, 28, 31, 32). Nonsynonymous variants are mostly weakly



**Box 1. Explanation of the main symbols used in the text**

$s_i$ is the selection coefficient for a mutation at the $i$th site under consideration; the ratio of the fitness of homozygous carriers of the mutation to the fitness of wild-type homozygotes is $1 - s_i$.

$h_i$ is the dominance coefficient for this mutation, such that the ratio of the fitness of heterozygotes to wild-type fitness is reduced below 1 by $t_i = h_i s_i$.

$\alpha_i$ measures the effect of a mutation at site $i$ on a fitness component $z$ relative to its effect on fitness, such that $\delta z_i = \alpha_i s_i$, where $\delta z_i$ is the homozygous effect of the mutation on $z$.

The mean, root mean square and coefficient of variation (ratio of the SD to the mean) of a variable $x_i$ (e.g., $s_i$) are denoted by $\bar{x}$, $\tilde{x}$ and $CV_x$, respectively; note that $(\tilde{x}/\bar{x})^2 = 1 + CV_x^2$.

The mean number of new mildly deleterious mutations per haploid genome per generation caused by base substitutions plus small insertions/deletions is $U_N$; the corresponding contribution from large insertions/deletion or transposable elements is $U_L$; their total is $U_D = U_N + U_L$.

$D_M$ and $V_M$ are the rate of decline in mean and rate of increase in variance per generation of a trait, respectively, caused by the accumulation of deleterious mutations in the absence of selection.

$V_H$ is the variance among a set of genotypes that are homozygous for a given *Drosophila* chromosome, extracted from a sample of individuals by balancer crosses.

$V_A$ and $V_D$ are the additive and dominance variances among individuals heterozygous for chromosomes extracted from a population (contributions from epistatic variance components have been ignored– see ref. 36 for a justification for this).

$\bar{v}$ and $CV_v$ are the mean and coefficient of variation of the effects on viability of new, mildly deleterious mutations that contribute to $U_L$.

$B$ is the inbred load of a trait $z$, defined as the difference between the natural logarithms of the mean value of $z$ for heterozygotes and homozygotes, for chromosomes extracted from a population by balancer crosses. The mean value per site contributing to the load is $\bar{B} = B/n$, where $n$ is the number of sites in question.

selected against, with a wide distribution of selection coefficients around the mean. Fewer than 10% of new mutations fall into the region where they are effectively neutral ($N_e t$ between 0 and 0.5). Little is known about the DFE for noncoding sequence mutations, which contribute the majority of sites under selective constraint (11, 33). However, their levels of polymorphism and divergence suggest that they are typically subject to much weaker selection than nonsynonymous mutations (11, 34), and therefore will contribute relatively little to the variances in fitness components that will be discussed next.

## Quantitative Genetics Analyses of Fitness Components in *Drosophila*

The other biological level for which plentiful data on variability are available is that of quantitative traits. Here, I will consider evidence concerning traits related to fitness. Although we would ideally like to have data on the genetics of net fitness, these are hard to obtain for diploid genotypes (35). Most quantitative genetics studies of *Drosophila* therefore use fitness components, measured under laboratory conditions. A fitness component is formally defined as a trait for which an increase in its value is associated with an increase in net Darwinian fitness, if the values of other traits are held constant (36). Inferences concerning the causes of variation in fitness components can be obtained by combining (*i*) information on the rate at which variability in fitness components accumulates by mutation with (*ii*) statistical analyses of natural variation (36–41).

Studies of this kind have frequently made use of balancer chromosomes to study the effects of one of the major chromosomes in the *Drosophila* genome. A balancer carries a set of inversions that suppress crossing over along the length of the chromosome in question, when heterozygous with a normal chromosome (42, 43). Deleterious mutations arising on this chromosome (other than dominant lethals or steriles) can then be sheltered from selection in MA experiments, by backcrossing males that are heterozygous for the balancer and a wild-type (WT) version of the chromosome to females from the balancer stock. Independent WT chromosomes can also be extracted from natural populations, and the trait values of individuals homozygous for such chromosomes can be measured. By intercrossing sets of independently extracted WT chromosomes, a "diallel cross" can be made, allowing estimates of components of genetic variance to be obtained for outbred individuals that are representative of genotypes from the original population, as well as for chromosomal homozygotes (36, 40, 44, 45).

By combining information from both types of study, we can ask questions about the strength of selection acting on deleterious mutations and compare the answers to those from the population genomics studies described above. We can also ask whether the levels of variation among genotypes taken from a wild population can be explained purely in terms of mutation selection balance or whether other processes such as balancing selection make significant contributions.

**Some Basic Models.** The inferences by which answers to these questions can be obtained are based on simple quantitative genetics models. These models can be introduced by the following method for estimating $\bar{t}$ (36), which compares the variation in a fitness component among homozygotes for chromosomes extracted from a population ($V_H$) with the rate at which variation in this trait arises by mutation ($V_M$), on the assumption that $V_H$ results entirely from mutation selection balance. The fitness of a heterozygous carrier of a mutation at a given nucleotide site $i$ is assumed to be reduced by $t_i = h_i s_i$, where $s_i$ is the homozyous selection coefficient for the mutation in question, and $h_i$ is the dominance coefficient (Box 1). Much evidence suggests that that $h_i$ for deleterious mutations is mostly nonzero but <0.5, i.e., mutations are partially recessive in their effects on fitness (1, 36, 37, 46–48).

To analyze the experimental results on fitness components, we need a model that describes how the effects of mutations on a given fitness component are related to their effects on fitness itself. For a trait value represented by a variable $z$, let the homozygous effect on $z$ of a mutation at a given nucleotide site $i$ (relative to the WT mean) be $\delta z_i = \alpha_i\, s_i$, where $\alpha_i$ represents the effect of the mutation on the trait, measured relative to its effect on fitness. The heterozygous effect of the mutation on $z$ is then $\alpha_i\, h_i\, s_i = \alpha_i\, t_i$ (36).

This model leads to an expression for $V_M$. For simplicity, $\alpha_i$ and $s_i$ are assumed to be independent. An investigation of the sensitivity of the results to violations of this assumption is presented in *SI Text* and Tables S1–S6; this shows that the main conclusions presented below are robust to such violations. Eq. **S1** implies that

$$V_M = U_D \tilde{\alpha}^2\, \tilde{s}^2, \qquad [1]$$

where $U_D$ is the deleterious mutation rate; and $\tilde{\alpha}$ and $\tilde{s}$ are the root mean squares of $\alpha_i$ and $s_i$, respectively, over all mutable sites that contribute to variation in fitness (mutations that do not affect the trait in questions have $\alpha_i = 0$).

An expression can also be obtained for $V_H$, under the assumption that variability arises solely from the balance between mutation selection. In this case, $z$ is scaled relative to the mean value over all lines to provide a dimension-free measure of variability (39, 49). (This procedure assumes that the effects of variants at different sites are additive on a logarithmic scale, but this is nearly equivalent to additivity on the original scale if the individual effects of variants are small.)

As shown in *SI Text*, we have

$$\frac{V_M}{V_H} \approx \bar{t}\left(1 + CV_t^2\right), \qquad [2]$$

($CV_s$ should really be used here, but Eq. **S8** shows that $CV_t \approx CV_s$.)

Given an estimate of $CV_t$ (see the above discussion of polymorphism data), we can then compare values of $\bar{t}$ obtained from the population genomics analyses to the result of applying Eq. **2** to quantitative genetics data.

**Some Relevant Quantitative Genetics Data.** As far as $V_H$ for *D. melanogaster* is concerned, the largest datasets come from studies of egg-to-adult viability, especially for the second chromosome. This chromosome represents about 37% of the sequenced euchromatic genome (50), so that estimates of genetic variances for the second chromosome are likely to be less than half that for the genome as a whole. $V_H$ for the second chromosome can be estimated from crosses that use balancer chromosomes to extract large numbers of WT chromosomes from natural populations, yielding good statistical precision. A compilation of estimates is given in Table S7. The "quasi-normal" class of chromosomes (those with viabilities >50% of the mean viability of chromosomal heterozygotes) contains small effect variants that are most comparable with those analyzed by the population genomics methods, so this class will be the main focus of attention here. For these chromosomes, $V_H$ for viabilities normalized by the mean for chromosomal homozygotes spans a range of ~0.01–0.04.

There is substantial variation in estimates of $V_M$ for the second chromosome among different MA experiments (51). For comparison with the estimate of $V_H$, data on quasi-normal chromosomes are used here, with a consensus value of $V_M = 10^{-4}$ after normalization of viabilities by the mean viability for the initial generation (*SI Text*). This procedure gives $V_M/V_H = 0.010$ with $V_H = 0.01$, and 0.0022 with the highest estimate of $V_H$ in Table S7.

As described above, the population genomics approach yielded an estimate of $\bar{t}$ of 0.0010 for nonsynonymous mutations, with a coefficient of variation of 2.0 (29). If this estimate of $CV_t$ is used in Eq. **2**, the estimates of $\bar{t}$ from the quantitative genetics data from the above two values of $V_M/V_H$ are 0.0020 and 0.00044, respectively, in fairly good agreement with the population genomics estimate. However, we also need to ask if the magnitudes of $V_M$ and/or $V_H$ are explicable on the basis of this value of $\bar{t}$, which can be done for $V_M$ as follows ($V_H$ will be considered later).

From the results in *SI Text*, we have $\bar{t} \approx \bar{h}\bar{s}$, where $\bar{h}$ is the mean dominance coefficient. Substituting this into Eq. **1**, we obtain

$$V_M \approx U_D \tilde{\alpha}^2 \ \bar{t}^2 \left(1 + CV_s^2\right) \Big/ \bar{h}^2. \qquad \textbf{[3]}$$

Estimates of $\bar{\alpha}$ and $\tilde{\alpha}^2$ can be obtained as described in *SI Text*; Table S8 summarizes available estimates for *D. melanogaster* fitness components. Although the values for individual traits have wide confidence intervals, the general pattern of positive values that are all <1 is very clear.

With $U_D = 0.185$ for the second chromosome, $CV_s = CV_t = 2$ and $\bar{t} = 0.002$, the numerator of the right side of Eq. **3** becomes $0.185 \times (0.002)^2 \times 5 \times \tilde{\alpha}^2 = 3.70 \times 10^{-6} \times \tilde{\alpha}^2$. If the largest estimate of $\tilde{\alpha}$ for viability in Table S8 is used (0.69), $\bar{h} = 0.13$ is required for $V_M = 1 \times 10^{-4}$; if the population genomics estimate of $\bar{t} = 0.001$ is used, we need $\bar{h} = 0.066$. Even lower values are needed with smaller $\tilde{\alpha}$ values. Such low $\bar{h}$ values are inconsistent with the fact that $\bar{h}$ for mildly deleterious mutations detected in MA experiments is ~0.25 (37, 47, 48) (see below for an additional estimate). There is thus a major discrepancy between the value of $V_M$ and the population genomics estimate of $\bar{t}$. A similar conclusion was reached in ref. 7, using a somewhat different argument.

**The Mutational Decline in Mean Viability.** A similar problem arises with respect to estimates of $D_M$, which is the rate of decline per generation of the mean of a set of MA lines. Unfortunately, estimates of $D_M$ for viability show considerable variation among different MA experiments (51–53), which has led to a good deal of disagreement about the interpretation of MA experiments. However, as described in *SI Text*, the more recent experiments, and some reanalyses of the older experiments, suggest a consensus value of $D_M$ for the second chromosome of ~0.002.

$D_M$ can be interpreted using the same model as above, leading to the expression

$$D_M = U_D \bar{\alpha} \ \bar{s}, \qquad \textbf{[4a]}$$

where $\bar{\alpha}$ and $\bar{s}$ are the arithmetic means of $\alpha$ and $s$, respectively.

Substituting from Eq. **4a** into Eq. **1** gives

$$\frac{V_M}{(D_M)^2} = \frac{\tilde{\alpha}^2 \left(1 + CV_s^2\right)}{\bar{\alpha}^2 U_D}. \qquad \textbf{[4b]}$$

Note that $\tilde{\alpha}^2/\bar{\alpha}^2 = 1 + CV_\alpha^2$ (Box 1), so that $\tilde{\alpha}/\bar{\alpha} \geq 1$. Given an estimate of $\tilde{\alpha}/\bar{\alpha}$ and the other variables in Eq. **4b**, we can estimate $CV_s$.

Using the above value of 0.5 for the deleterious mutation rate for nucleotide substitutions and small indels and 0.37 for the proportion of the euchromatic genome represented by the second chromosome, the deleterious mutation rate for this chromosome is $\sim 0.37 \times 0.5 = 0.185$. However, this may underestimate $U_D$, because it does not take into account sources of mutation such as transposable element insertions (see below). If we ignore this difficulty for the moment, and set $U_D = 0.185$ for the second chromosome, use of the lower bound of one for $\tilde{\alpha}/\bar{\alpha}$, together with the above values of $D_M$ and $V_M$ (which give $V_M/D_M^2 = 25$), yields $CV_s = 1.9$, slightly less than the population genomics estimate of $CV_t$.

However, again we need to ask whether the magnitude of $D_M$ is consistent with these estimates. From Eq. **4a** and the argument following Eq. **S7**, we have

$$D_M \approx U_D \bar{\alpha} \ \bar{t} \big/ \bar{h}. \qquad \textbf{[5]}$$

With $\bar{\alpha} = 0.34$ for viability (the highest estimate in Table S7), $\bar{t} = 0.0017$, and $U_D = 0.185$, this gives $\bar{h} = (0.0017 \times 0.185 \times 0.34)/(0.002) = 0.053$, which is in even worse agreement with the probable true value of $\bar{h}$ than the result for $V_M$.

**Reconciling the Two Approaches.** The most likely explanation for these inconsistencies between the population genomics and quantitative genetics approaches is that mutations with effects of the sizes estimated from the population genomics data are not the only contributors to $D_M$ and $V_M$. There must be an additional source of mildly deleterious mutations, probably mainly involving transposable element (TE) insertions (7, 54) and other types of large indels (12). These types of mutation are likely to have much larger effects on viability of their heterozygous carriers than most single nucleotide mutations or small indels, because they can completely disrupt sequences of functional importance; a minority of single nucleotide changes may also contribute to this class. This expectation is supported by experimental results (12, 55–57). The possible contribution of TEs to this class of mutation is assessed in *SI Text*.

The consequences of this contribution can be modeled as follows. Assume that there is a set of quasi-normal mutations with relatively large effects, arising at a net haploid mutation rate of $U_L$, and with a mean effect on viability of $\bar{v}$, so that $U_D = U_N + U_L$, where $U_N = 0.185$ is the mutation rate for nucleotide substitutions and small indels. Combining the two classes of mutation yields a simple generalization of the standard Bateman-Mukai relations (58, 59) for $D_M$ and $V_M$ (Eq. **S9**). Using the estimates of $U_N$, $D_M$, and $V_M$, these equations yield $U_L/(1 + CV_v^2) = 0.032$ and $\bar{v}(1 + CV_v^2) = 0.055$.

This result agrees with previous suggestions that mutations with very small fitness effects do not contribute significantly to $D_M$ and $V_M$, so that the quasi-normal chromosomes that appear in MA experiments are caused by mutations arising at a relatively low rate, but with substantial homozygous effects on viability, as high as 0.055 or more (53, 54, 60). It is also consistent with the observation that the distribution of viability among chromosomes subjected to EMS mutagenesis (61), which induces base substitutions, shows a very different pattern from that seen in MA experiments, with a dearth of quasi-normal chromosomes (54). This pattern is what is expected if TE insertions or large indels with moderate effects on viability are the main source of quasi-normal chromosomes, with point mutations mostly having much smaller effects.

**The Causes of Natural Variation in Fitness Components.** We can now ask what light the quantitative genetics approach has shed on the sources of natural variation in these traits. First, positive mutational correlations are consistently found between different fitness components (62–64). Corresponding to this, the overall effects of deleterious mutations on any single component, as measured by the $\bar{\alpha}$ and $\tilde{\alpha}$ parameters described above (Table S8), are often only a fraction of their net fitness effects (37, 41, 62, 65–67). Similarly, the net fitness reduction caused by homozygosity for second or third chromosomes extracted from natural populations is several times the corresponding reduction for viability (37, 68). These findings imply that deleterious mutations often affect more than one fitness component, with effects that are in the same direction.

Second, we can use these findings to make predictions about the levels of standing variation expected under the mutation selection balance hypothesis, using the estimates of $\bar{\alpha}$ and $\tilde{\alpha}$, as well as $D_M$ and $V_M$. The most extensive data against which these predictions can be tested are for the additive genetic variance in viability for chromosomal heterozygotes ($V_A$), which has been estimated in several diallel cross experiments on second chromosomes extracted from natural populations (40). The predicted value of $V_A$ under mutation selection balance is approximately $2U_D\tilde{\alpha}^2\bar{t}$ (36). Using Eq. **5** to eliminate $U_D$, we find that

$$V_A \approx 2\ \tilde{\alpha}^2\,\bar{h}\,D_M/\bar{\alpha}. \qquad [6]$$

Because the relevant diallel cross experiments did not separate chromosomes according to their homozygous fitness effects, it is necessary to include contributions from lethals in the predictions; severely detrimental mutations appear to be sufficiently rare that they barely affect $D_M$ estimates, so that the quasi-normal rates will be used for the nonlethal class. The second chromosome lethal mutation rate is ~0.01, and heterozygotes for a lethal and a nonlethal chromosome appear to suffer a viability loss of $t_l$ = 0.01–0.02 (37). Using the higher value of $t_l$, the contribution to $V_A$ is $2 \times 0.02 \times 0.01 \times \tilde{\alpha}^2/\bar{\alpha} = 0.0004\ \tilde{\alpha}^2/\bar{\alpha}$. With the previously used (and somewhat conservative) estimates of $\bar{\alpha} = 0.34$ and $\tilde{\alpha} = 0.69$, this gives a contribution to $V_A$ of 0.00056. The highest reported estimate of $D_M$ for quasi-normals for the second chromosome is from the original experiment of Mukai (59); a reanalysis of these data gave a value of $D_M = 0.0055$ (69). With $\bar{h} = 0.25$, the predicted $V_A$ contributed by nonlethals is then 0.0035, giving a total of 0.0041, mostly coming from the nonlethal chromosomes. With the consensus $DM$ = 0.002, the prediction is $V_A$ = 0.0028.

Estimates of $V_A$ vary considerably [see refs. 38 and 40 (table 1)]. Two northern Japanese populations had $V_A$ values for viability that are close to the mutation selection balance prediction. In contrast, $V_A$ estimates from US and southern Japanese populations were all much higher than the above prediction, with values as high as 0.02 and a mean of 0.015 ± 0.003. A possible explanation for the small values for the northern Japanese populations is that these have experienced recent bottlenecks of population size, leading to depleted variability; there is abundant evidence for such bottlenecks in non-African populations (70, 71). It seems clear, however, that forces other than mutation selection balance must be important for maintaining variation in the other populations. A similar conclusion is reached from consideration of the ratios $V_M/V_A$ and $V_M/V_H$ (*SI Text*).

**Some Further Implications.** These analyses suggest that there is too much genetic variability in fitness in many populations of *D. melanogaster* to be explained by the mutation selection balance hypothesis. In addition, the bulk of the variance that is contributed by mutation selection balance comes from the mildly deleterious mutations with relatively large effects; with the parameter estimates obtained from Eq. **S9**, these are expected to contribute 88% of the net $D_M$ for quasi-normal chromosomes. This quantity also represents their proportional contribution to the mutational components of $V_H$ and $V_A$ for this class of chromosome.

The first of these conclusions casts doubt on the frequently used method of estimating the mean of $h_i$ from the regression of the viability of chromosomal heterozygotes on the mean viability of homozygotes for the two parental chromosomes (36, 37, 72), because this assumes mutation selection balance. The same applies to the use of $V_M/V_A$ or $V_M/V_H$ to estimate the mean selection coefficient (see above) (73, 74).

The second conclusion means that, even if we could safely assume that the mutational contribution to the variances predominate, inferences from quantitative genetics apply mainly to the large effect class of mutations and tell us little about the minor effect mutations that probably make up the bulk of spontaneous mutations that affect fitness and that are detected with the population genomics approach. This problem can be illustrated by examining the use of the ratio of the inbred load to $D_M$ as an estimate of $\bar{t}$ for a mildly deleterious mutation, proposed by Crow and Simmons (37). For balancer experiments, the inbred load, $B$, for a trait is defined as the difference between the natural logarithms of the mean value of $z$ for chromosomal heterozygotes and homozygotes (75); for small values of $B$, this is approximately the same as the difference between the raw means, divided by the raw mean for the heterozygotes. This method has generated the widely quoted result that $\bar{t}$ is ~2% (36, 37, 41, 74), which is of course inconsistent with the population genomics estimate of 0.1%.

The analysis described in *SI Text* uses information on $B$ to confirm the view that mildly deleterious mutations are far from completely recessive with regard to their fitness effects (1), as is also suggested by studies of the effects of heterozygous new mutations on fitness components (37, 41, 47, 48). This analysis also yields an estimate of $\bar{t}$ of 0.044, which is much larger than the Crow-Simmons estimate, and would be even larger if mutation selection balance were not the only source of the inbred load.

## Discussion

The above results suggest that the mean selective effect of a new, mildly deleterious mutation in *Drosophila* is greatly underestimated by recent analyses of DNA sequence polymorphism data (29, 30). These analyses are based on variants that are found in relatively small samples from the population; this means that the contributions to the DFE from mutations that are so strongly selected that they do not appear in the sample are not taken into account (23, 24). There is thus no reason to be alarmed about this discrepancy. The estimates of the mutation rates for the two classes of mutation suggests that mutations with relatively large effects represent only a small fraction of the total input of mildly deleterious mutations. Nonetheless, their net contribution to $\bar{t}$ and to the standing genetic variance caused by mutation-selection balance outweighs the contribution from the more numerous mutations with much smaller effects.

Seemingly paradoxically, DFE studies of humans suggest $\bar{t}$ values of 0.02 or more (24, 26, 27), over an order of magnitude greater than the *Drosophila* value. A possible explanation for this is the much smaller $N_e$ of humans compared with *Drosophila*; the probability distribution of variant frequencies is determined by $N_e t$, not $t$, so that large effect variants contribute more to human polymorphism datasets. Taxa like *Drosophila*, with large effective population sizes, may thus not be the most suitable material for estimating the DFE from population genomic data.

Ironically, the classic *Drosophila* value of $\bar{t}$ between 0.01 and 0.02 (37) seems to be closer to the truth than the population genomics estimates, although even this is likely to be a substantial underestimate (see above). This result has implications for endeavors such as attempts to fit models of the effects of deleterious mutations on variability at linked sites to observed patterns of variability across the genome (76, 77); use of the population genomics estimate of the DFE could underestimate these effects, especially for regions that are some distance away from the targets of selection, because more strongly deleterious mutations have effects that extend over longer genetic distances than those of weakly selected mutations.

An important implication of the results described here is that they reject the hypothesis that variability in fitness components in *Drosophila* populations is maintained solely by a balance between the mutational input of deleterious variants and their elimination by selection. Instead, much of this variation appears to reflect the effects of some form of balancing selection or an interaction between migration and spatial variation in selection pressures. Detailed models of the relevant processes have been widely discussed in the population genetics literature (78). Favorable mutations that are in the process of spreading to fixation could also contribute to variation in fitness; however, the analysis presented in *SI Text* suggests that these are unlikely to be important.

An interesting observation is the fact that the genetic variance in viability for the small fourth chromosome of *D. melanogaster* appears to be approximately one-half of that for the second chromosome (79), despite the fact that it contains less than 1/20th the number of genes. This finding strongly suggests the maintenance of variability by balancing selection; intriguingly, DNA sequence data indicate the existence of two major haplogroups on the fourth chromosome, despite its very low overall level of variability (80, 81). It would be worth examining whether these haplogroups are associated with different viabilities of their carriers.

Further evidence for a major role for balancing selection in contributing to genetic variance in fitness components is provided by a test due to Kelly (82). This test uses the theoretical result that the relative magnitudes of the effects of artificial selection on the population mean and inbreeding depression for the selected trait can distinguish between variation maintained primarily by mutation-selection balance and variation with a substantial contribution from alleles at intermediate frequencies (implying balancing selection). When applied to a selection experiment on female fecundity in *D. melanogaster*, the test suggested an important role for intermediate allele frequencies (83), as did selection experiments on several traits in the monkeyflower *Mimulus guttatus* (84, 85). This method could usefully be applied more widely.

A major contribution to these high levels of variability from such factors as inversion polymorphisms and the selective effects of heterogeneous environments (38, 40) can probably be ruled out by the following argument. Most of the data on fitness components other than viability (36, 45, 86), whose genetic variances are similar to those for viability, came from a single laboratory population of *D. melanogaster*, *IV*, which was founded from wild-caught flies in 1975 and subsequently rendered free of inversion polymorphisms (87). The selection experiment on fecundity just mentioned also used this population. A similarly high variance in the net fitness of flies that were heterozygous for WT third chromosomes and a balancer chromosome was estimated for another long-term laboratory population (35). It is, in fact, remarkable how much genetic variability in quantitative traits is maintained in small closed populations (88).

The conclusion that much variation in viability and other fitness components in many *Drosophila* populations is maintained by forces other than mutation selection balance is not new—it was arrived at in the 1980s (38–40). However, the basis for it is now much stronger, especially when the information from direct measurements of mutation rates is taken into account. This conclusion should not be taken to mean that balancing selection is ubiquitous across the genome. Population genomics results provide little evidence for balancing selection, suggesting only a scattering of loci across the genome with alleles that have been maintained at intermediate frequencies by selection for periods of time that are much larger than the mean neutral coalescence time (77, 89–91). However, it is well known that such loci contribute disproportionately to the genetic variance, so that a relatively small number could cause most of it, even if the vast majority of individual variants affecting fitness were kept at low frequencies by purifying selection (5). The inferences from population genomics studies and quantitative genetics do not, therefore, contradict each other.

There remains, however, a technical question arising from the quantitative genetics results. The diallel cross experiments on viability have repeatedly revealed abundant additive genetic variance in southern US and Japanese populations of *D. melanogaster*, but very little dominance variance (38, 40). If variation is mainly caused by some form of balancing selection, this observation appears to contradict the very strong theoretical result that there should be no additive variance in fitness in a population at equilibrium under selection alone, a corollary of Fisher's fundamental theorem of natural selection (39, 92). It certainly excludes the possibility that there is often heterozygote advantage at the level of viability itself (36, 40).

Additive variance in fitness components can be maintained in the absence of $V_A$ for net fitness by processes such as frequency-dependent selection or antagonistic effects of alleles on different fitness components, given suitable conditions on the dominance coefficients for the effects of the variants on the traits in question (36). These processes would generally be expected to produce some dominance variance, $V_D$. The size of $V_D$ depends on the number of sites contributing to the inbred load ($n$), and the mean size of the inbred load contributed by each site ($\overline{B}$), in a way which is independent of the nature of the forces maintaining variability (36, 44) (see *SI Text* for details).

A compilation of the results of experiments on second chromosome effects on viability in *Drosophila*, where both components of genetic variances and $B$ were estimated, yields a mean value of $V_D$ of $0.0010 \pm 0.0002$ and $B = 0.33 \pm 0.04$ for all types of nonlethal chromosomes from US and southern Japanese populations, where $V_A$ values are high (data from table 1 of ref. 40). These estimates imply $n \geq 109$ and $\overline{B} \leq 0.003$, using Eq. **S13**. A relatively modest number of loci, each contributing a small inbred load in viability, is thus consistent with these results.

However, is this low value of $V_D$ consistent with the high mean value of $V_A$ ($0.015 \pm 0.003$), with $V_D/V_A$ equal to $0.070 \pm 0.013$? If variation were caused by biallelic loci with allele frequencies of one-half and the same effect sizes at each locus, the expected value of $V_D/V_A$ can easily be found (*SI Text*); to account for the observed value, the dominance coefficient must be equal to 0.32, which is perfectly plausible. This example is artificial, but it suffices to show that the relative values of dominance and additive variances for viability can be explained without difficulty.

In fact, viability seems to be the fitness component that exhibits the lowest value of $V_D/V_A$ in *D. melanogaster* (36). Some traits, such as sperm precedence, show no significant $V_A$ but significant $V_D$ and $B$, suggesting that variation is contributed by a few loci with major effects (45, 86). As was pointed out some time ago (86), these results imply that a search for polymorphisms with alleles with major effects on fitness components should be fruitful, especially when there is evidence for dominance variance. Modern genomic methods are producing findings that are consistent with this expectation (93–95).

If these conclusions are correct, what is the prospect for extending them to species other than *D. melanogaster*? The tests for nonmutational contributions to variation in fitness components described here depend heavily on estimates of $D_M$ and $V_M$ from MA experiments; such experiments in other higher organisms have mainly come from highly self-fertilizing species such as *Arabidopsis thaliana* and *Caenorhabditis elegans* (18), with one or two exceptions (96). The interpretation of variance components for natural populations of highly selfing or asexual species is problematical, because of the high levels of linkage disequilibrium associated with their low effective rates of genetic recombination and the likelihood of perturbing effects of hitchhiking events on allele frequencies (78); these violate the assumptions used here. Reliable estimates of genetic variances and inbred loads for additional outbreeding organisms, as well as $D_M$ and $V_M$ from MA experiments, would be of considerable interest.

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